Script Notes and Developer Log

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Contents

0.1 Overview

This document collects implementation notes, design decisions, and development comments across scripts used in the regional skew estimation project. It supports ongoing improvements and serves as a lightweight developer log.

Use this log to track:

- Known issues or limitations
- Planned improvements (`TODO`)
- Script-specific decisions or reminders
- Ideas for refactoring or cleanup

0.2 Project Notes (General)

0.3 Notes by Script

0.3.1 `01a_download_ecoregions.R`

- Original metadata & layer files for each level are downloaded for reference.
- Data sources are EPA/CEC shapefiles hosted via AWS links.
- This script assumes internet access and local write permissions.

0.3.2 01b__download__USGS__gage__data.R

- Requires internet access to download data from USGS NWIS
- Bounding box grid helps avoid request size limitations in NWIS queries
- Uses a batch download approach for peak flow data retrieval
- Great Plains extent is defined using EPA Level 1 Ecoregion shapefiles

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0.3.3 01d_download_us_peakflow_data.R

- The script inputs data from `01c_download_usgs_gage_metadata.R`
- The script does not apply filtering on regulation, dam failure, or record length.
- Filtering occurs in the subsequent script – `01e_filter_usgs_peakflow_data.R`

0.3.4 01e_filter_usgs_peakflow_data.R

- Bulletin 17C explicitly recommends using the Multiple Grubbs-Beck Test (MGBT) to identify low-end outliers, e.g. zero flows, and near-zero flows like 0.01, 0.02, 0.05, etc. MGBT() flags observations below a calculated threshold discharge.
- The inclusion of outliers (note in FFA there are only low-end outliers) distorts the estimation of the log-Pearson Type III distribution, biasing skewness, mean, and standard deviation. PILFs tend to skew the lower tail, by underestimating the sample skewness and producing unrealistic estimates for rare floods.
- Peak codes are comma-separated and must be parsed for reliable flagging
- Skewness is calculated as type = 1, using a method of moments estimator which follows the guidances under Bulletin 17C
- Install peakfq directly from USGS GitLab `remotes::install_gitlab("water/peakfqr", host = "code.usgs.gov")`

0.3.5 01f_download_nhdplus_v2.R

- The code retrieves flowlines and catchments intersecting the buffered AOI
- `get_nhdplus()` loops through 144 tiles that intersect AOI.
  - It fetches data chunk-by-chunk and begins stitching them together.
  - Midway or After Completion: Invalid Geometry Detected
  - The function detects one or more invalid geometries (e.g., self-intersecting polygons or degenerate line segments).
  - It triggers a geometry repair step internally – sf / s2 Geometry Repair Mode Kicks In. This temporarily activates s2 geometry engine to fix topological errors, which is common in large hydrologic datasets. You see:
    * Found invalid geometry, attempting to fix.
    * Spherical geometry (s2) switched on
    * Spherical geometry (s2) switched off
    * Tiles Are downloaded again. The function starts over, reloading the same set of tiles (tiles 1–144) with geometry corrections in place.
  - This is expected behavior in nhdplusTools when:
    * An invalid feature is encountered (common with complex catchments or clipped geometries), and get_nhdplus() must retry with geometry repair enabled.

0.3.6 01g_download_nhdplus_hr.R

Key Features of the script: - Idempotent: skips previously downloaded regions - Geometry-safe: cast, buffer, validate for AOIs - Fuzzy retry logic: name correction and tile fallback - Transparent logging: CSV-based log with timestamped status - Interactive QA: mapview preview of results

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0.3.7 01m_download_nlcd_2016.R

The NLCD raster was manually downloaded. Using {FdData} All tile requests timed out at ~30 seconds. Received over 100–200 MB per tile, but not enough to complete the download. Then FedData::get_nlcd(),tried to crop those partial files, which resulted in a crash.

The script inputs the following files and projects the raster to the vector CRS.

- a raster file: a SpatRaster (e.g., the full NLCD)
- a vector file: a SpatVector (e.g., Great Plains boundary)

Next, the script applies the following functions:

- crop() – Trims the raster extent down to the bounding box of the vector geometry (shape). Keeps all raster cells that intersect the box — even if they fall outside the exact shape.
- mask() – Replaces all raster cells outside the exact shape with NA. Keeps only values within the actual polygon boundary.

0.3.8 01n_download_ned.R

Slope was calculated using (Fleming & Hoffer / Ritter algorithms), which use only the four cardinal directions (rook's case) to produces smoother, more generalized slope surfaces, which better matches subtle terrain transitions, especially in agricultural, prairie, or floodplain contexts.

When clipping I used `expand = 1000` for buffer to help prevent clipping artifacts near edges

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0.3.9 01o_download_modis_ndvi_2016.R

NDVI, or Normalized Difference Vegetation Index is a measure of photosynthetic activity occurring within a pixel, which is commonly used to assess the health and density of vegetation. NDVI ranges from [-1, +1], with healthy vegetation typically between [0.6, 0.9]. Higher NDVI values indicate healthier, more vigorous vegetation.

- NDVI = (NIR - Red) / (NIR + Red)

NDVI for the Great Plains ecoregion is calculated from MODIS, or Moderate Resolution Imaging Spectroradiometer data on the NASA Terra satellite. The relevant data, 250m resolution, 16-day composites, are labeled as `MOD13Q1`.

- MOD = MODIS sensor on Terra satellite
- 13 = Product suite 13: Vegetation Indices
- Q1 = 16-day temporal composite
- 061 = Collection 6.1 (latest operational version)

MOD13Q1 tiles contain multiple subdatasets (SDS): - NDVI — Normalized Difference Vegetation Index - EVI — Enhanced Vegetation Index - VI Quality — Bit-packed QA layer - Reflectance — Red, NIR, Blue bands used for index calculation - Day of Year — Date of composite

MODIS data are projected to a sinusoidal projection to preserve the relative size of land areas on the Earth's surface. The sinusoidal projection is an equal-area projection. MODIS tiles are 10-degrees by 10-degrees at the equator, and are named using Cartesian coordinates such that a tile named like h10v05 indicates the tile is the tenth tile from the origin in the horizontal direction and the fifth tile in the vertical direction from the origin (see https://modis-land.gsfc.nasa.gov/MODLAND_grid.html).

NDVI was chosen over EVI or Enhanced vegetation index for the Great Plains ecoregion. EVI is an index similar to NDVI that reduces the effects of: canopy background, aerosols (blue-band correction), and soil brightness, which often improves performance in dense forests. NDVI was selected for this project for the following reasons:

- Moderate to sparse vegetation: prairie, cropland, rangeland.
- Low cloud/aerosol interference: no need for blue-band correction.
- Wide use in agriculture and rangeland applications, such as crop condition, forage productivity, and drought/grazing assessment (VegDRI, USDM, NDMC).
- Fewer assumptions and post-processing corrections.

0.3.10 02a__merge__nhdplus__hr__flowlines.R

Type coercion is the process of converting a value or object from one data type to another. This can occur either automatically (implicitly) or intentionally (explicitly). Explicitly fixes issues using a file-reading loop with type coercion and error logging. The loop fixes the following issues: - Only coerce if possible, using a safe test like is.integerish(). - Fallback to numeric when needed. - Applies coercion one column at a time - Skips columns that cause errors (and logs them) - Prevents across() from failing all at once - Wrap the across() call in logic that filters to only existing columns. - Fine-grained control: logs problems per column, per file - Non-blocking: does not interrupt the whole region's read if a single column fails - Quiet warnings: keeps logs readable but still lets you know what happened - coerce fdate to character safely in all files. Character is the most flexible, readable, and safest for uncertain timestamp formats.

0.4 TODO (General)

0.5 Ideas

0.6 Notes

- Last updated: 2025-07-25 17:24
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